OPPORTUNITY BRIEF · SALES AND PRESALES

Post-quantum cryptographic migration for financial services to counter harvest-now-decrypt-later threats, driven by NIST standards and enterprise readiness gaps.

Finance, Banking and Insurance × Post-quantum cryptography migration × Post-quantum cryptography · OS006 v4 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
72 is the world moving	**59** can we play, can we win	**MEDIUM** 1 named in the evidence	**€189m** modelled evidence · per year	**NEXT** regulation adopted or propose...	**L0** 25 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 4.0 published reference(s) in Finance, Banking and Insurance, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

Post-quantum cryptographic migration for financial services to counter harvest-now-decrypt-later threats, driven by NIST standards and enterprise readiness gaps. This opportunity targets banks and insurers that must protect long-lived data and transactions from future quantum decryption, and it exists now because NIST has finalized PQC standards while enterprise adoption remains critically low.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€231m** €41.6m – €1.1bn	**€189m** €33.9m – €907m	**€6.6m** €1.2m – €31.8m	modelled
Observed: tendered contract value, annualised	€100m	€100m	€3.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Finance, Banking and Insurance (EU27_2020)	11,116 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Post-quantum cryptography	39.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cisce_ran2 · E_SECMDENC	2024
Annual value of one engagement	€240k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	2,426 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. ICT security measures by NACE Rev. 2 activity series E_SECMDENC is used as a proxy for Post-quantum cryptography: Encryption users are the migration population. Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract value is a configured assumption, not observed. Only 2 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 3 declared geographies are outside the European reference data (US, IN); the estimate covers the European footprint only. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes the financial sector; PROXY — auxiliary financial activities; PROXY — insurance and pension funding

Observed: tendered contract value, annualised

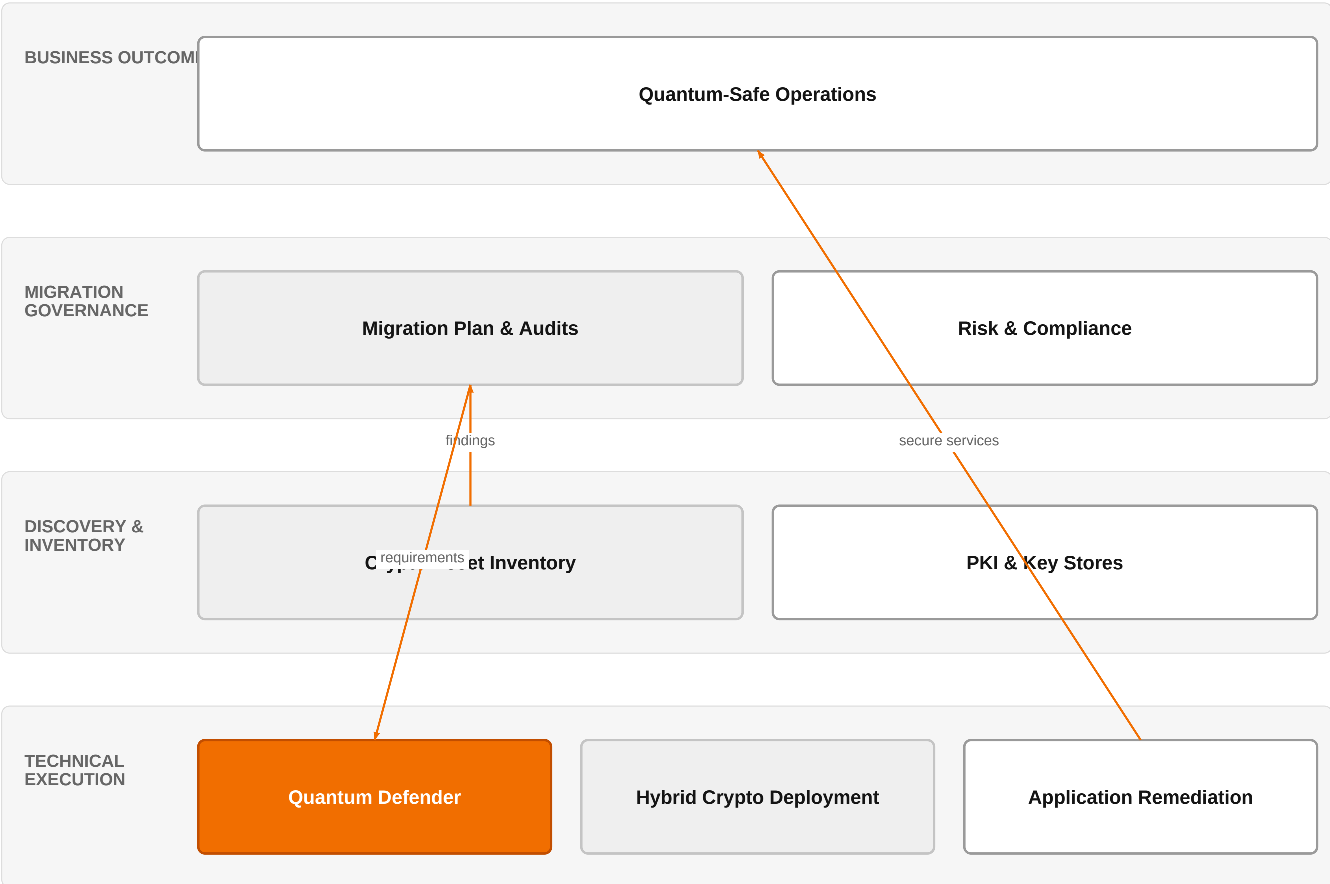
Factor	Value	Basis	Source	As of

Tendered contract value in matching CPV categories	€100m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 3 declared geographies are outside the European reference data (US, IN); the estimate covers the European footprint only.

The solution, in one picture

PQC Migration for Financial Services



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's Quantum Defender and migration expertise integrate with the customer's existing systems to achieve quantum-safe operations.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

NIST has finalized post-quantum cryptography standards (FIPS 203, 204, 205), initiating an unprecedented cryptographic migration, yet enterprise adoption remains critically low. Adversaries can record encrypted traffic today and decrypt it later, a tactic known as harvest-now-decrypt-later, which is a temporal risk for financial data with long confidentiality requirements. Relevant guidelines are setting migration deadlines just a few years in the future, and PQC migration is a multi-year, multi-domain transformation across cloud, enterprise, embedded, and operational technology systems.

`Sources: SIG-B5F1E5C72CAB Springer Science and B 2026-07-30; SIG-40B64E273856 International Journal 2026-07-30; SIG-C822609B6A07 openalex.org 2026-05-31; SIG-57F4F0DAE044 Cryptography 2026-06-18`

Who buys, and why now

The Chief Information Security Officer (CISO) signs, but the pain is felt by the Chief Risk Officer and the second line of defense, who must ensure the migration is auditable and aligned with regulator expectations. The trigger is the finalization of NIST standards and the recognition that harvest-now-decrypt-later exposes long-lived financial data and transactions. The pain is operational: enterprises must migrate thousands of applications relying on cryptographic primitives, each requiring distinct remediation strategies and varying effort, and PKI migration is complex because these systems serve millions of users.

Sources: SIG-5C7568EBD975 Elsevier BV 2026-01-01; SIG-CE5A95F7AF72 openalex.org 2026-01-06; SIG-C822609B6A07 openalex.org 2026-05-31

Why it is hot now — every claim bound to a dated source

- Large-scale quantum computers threaten public-key cryptography that protects enterprise data, communications, and digital identity. [SIG-B8561992126B]
- Adversaries can record encrypted traffic today and decrypt it later, a tactic known as harvest-now-decrypt-later. [SIG-40B64E273856, SIG-B8561992126B, SIG-CA65D28B8AC6]
- Migration of Public Key Infrastructures poses specific challenges, as these systems are complex, serving millions of users and fundamental in providing cybersecurity. [SIG-C822609B6A07]
- Enterprises operating large-scale networks must migrate thousands of applications that rely on cryptographic primitives, each requiring distinct remediation strategies and varying effort. [SIG-CE5A95F7AF72]
- Federal post-quantum cryptography migration is scoped around three categories of cryptographic assets: libraries, protocols, and key stores. [SIG-444983B4C996]
- Cryptographic functions and key material can be realized in the parameters of machine learning models, and current open-source serialization-focused scanners we evaluated do not detect them. [SIG-444983B4C996]
- The advent of quantum computers poses a fundamental threat to current cryptography-based applications. [SIG-B8561992126B]
- We present a topical survey and comparative analysis of the transition strategies adopted by leading cybersecurity agencies—including NIST, ANSSI, and BSI. [SIG-353070727CBB]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a structured migration engagement leveraging its Quantum Defender / quantum-safe offers, supported by Cybersecurity experts and Orange Group researchers. The engagement would start with a crypto discovery and inventory phase to identify cryptographic assets across libraries, protocols, and key stores, then move to crypto-agility decisions, hybrid deployment sizing, PKI impact modeling, and finally ticket-ready execution waves. This approach is grounded in the PQC Migration & Crypto Inventory Lab methodology, adapted to the customer's environment.

Why Orange can win

Orange can win because it combines a quantum-safe offer (Quantum Defender) with deep cybersecurity expertise and the credibility of Orange Group researchers. The reference to Banqsoft and Stø demonstrates prior work in financial services, and Orange's certifications (ISO 27001, SOC 2 Type II, CISPE) provide assurance. However, the assets are thin: the offer is at L0 (early stage) and there is no named reference specifically for PQC migration, so Orange must lean on its research and integration capabilities.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SOC 2 Type II	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Banqsoft	reference	SUP	published reference in this vertical, different use case	unconfirmed
Stø	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.4; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include Microsoft (also an Orange partner) which is strong in cloud and has been named in evidence on the role of the second line of defense; Atos/Eviden, IBM, and Accenture as global systems integrators with strong enterprise reach; and security specialists like Airbus Protect, Entrust, ID Quantique, and Thales, which have specific PQC products. The customer will compare Orange against these on the basis of end-to-end migration capability, trust, and sovereignty.

Sources: SIG-5C7568EBD975 Elsevier BV 2026-01-01

Competitor	Category	Position here	Basis	Evidence
Microsoft	Hyperscaler	named in this topic's evidence — also an Orange partner	evidenced	SIG-5C7568EBD975
Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; active in Finance, Banking and Insurance; competes in Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; active in Finance, Banking and Insurance; competes in Cybersecurity	structural	—
Accenture	Global systems integrator	active in Finance, Banking and Insurance	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. Have you inventoried your cryptographic assets across libraries, protocols, and key stores, and do you know which of your systems are exposed to harvest-now-decrypt-later?
2. What is your current timeline for migrating to NIST post-quantum standards, and who in your organization owns the migration plan?
3. Do you have any systems that must protect data for more than a decade, such as long-term records or pension data, that would be at risk from harvest-now-decrypt-later?
4. Have you assessed the impact of PQC migration on your real-time payment gateways or other latency-sensitive financial services?
5. Are you aware of any cryptographic material embedded in machine learning models or other non-traditional assets that current scanners might miss?
6. What is your current relationship with your second line of defense regarding cryptographic auditing and compliance expectations?

Objections you will meet

They will say	You can answer

We don't see quantum computers as an immediate threat, so why should we invest now?	It's true that a cryptographically relevant quantum computer does not exist yet, but adversaries can record encrypted traffic today and decrypt it later. For financial data with long confidentiality requirements, that is a real risk. NIST has already finalized standards, and guidelines are setting deadlines just a few years out, so starting now is prudent.
We already have a security team and tools; why do we need Orange?	Your team may have tools, but PQC migration is a multi-year, multi-domain transformation that requires crypto discovery, crypto-agility decisions, hybrid deployment sizing, PKI impact modeling, and execution waves. Orange brings a structured methodology and quantum-safe offers, backed by research and certifications, to complement your team's expertise.
We are considering a hyperscaler or a security specialist for this; what makes Orange different?	Hyperscalers and security specialists have strengths, but Orange combines a quantum-safe offer with cybersecurity expertise, research capabilities, and a track record in financial services. We also bring a sovereign, trusted approach that is important for regulated industries.

Risks and unknowns

The main risk is that enterprise adoption remains critically low, so customers may not yet feel urgency. There is also uncertainty about the exact timeline for a cryptographically relevant quantum computer, and the migration is complex and multi-year, requiring ecosystem coordination. Additionally, current tools may miss cryptographic material embedded in machine learning model parameters, creating unknown exposure. Orange's offer is at an early stage, so delivery capability is not yet proven at scale.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Bid brief: for a financial services client, assemble a proposal that differentiates our Quantum Defender offer by highlighting our PQC Migration & Crypto Inventory Lab, our cybersecurity experts, and our certifications (ISO 27001, SOC 2 Type II, CISPE), and reference Banqsoft and Stø as proof points.
Sales	In a conversation with a European bank's CISO, say: 'Given the NIST standards are finalized and harvest-now-decrypt-later is a real threat, we can help you move from knowing you need PQC to a concrete migration plan. Our Quantum Defender offer, backed by our ISO 27001 and SOC 2 Type II certifications, can support your crypto discovery and migration planning.'
Strategist / Innovator	Deep-dive brief: assess the enterprise readiness gap for PQC migration in financial services by mapping our Quantum Defender offer against the three categories of cryptographic assets (libraries, protocols, key stores) and the migration challenges highlighted in the evidence, to decide where to invest study and prototyping effort next quarter.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-40B64E273856	2026-07-30	International Journal of ...	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-B5F1E5C72CAB	2026-07-30	Springer Science and Busi...	1	Post-Quantum Cryptography Migration Readiness in Global Capability Centres: A Governance Framew...
SIG-B8561992126B	2026-07-30	openalex.org	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-57F4F0DAE044	2026-06-18	Cryptography	1	Mythos-Class Frontier Models and the Compression of Post-Quantum Cryptography Migration Timelin...
SIG-81E56EEBF9D4	2026-06-15	openalex.org	1	Post-Quantum Cryptography Migration for Agentic AI Systems
SIG-C822609B6A07	2026-05-31	openalex.org	1	Large Scale Actualisation of Post-Quantum Cryptography Migration
SIG-444983B4C996	2026-05-09	openalex.org	1	Parameter-Resident Cryptographic Material as an Unscoped Surface for Post-Quantum Migration: An...
SIG-BFFA5CE48BF5	2026-02-27	Shanlax International Jou...	1	Migration Strategies from Traditional Cryptography to Post-Quantum Cryptography in Cloud Systems
SIG-964C3B0C145D	2026-02-19	openalex.org	1	Post-Quantum Cryptography - Migration & Crypto Inventory Lab
SIG-969C5FDB15DD	2026-02-02	Institute of Electrical a...	1	Post-Quantum Cryptography Migration Framework for Real-Time Payment Gateways
SIG-13AAE2CD702E	2026-01-19	MDPI AG	1	Cryptographic Asset Discovery and Inventory for Embedded Systems: A Framework for Post-Quantum ...
SIG-53F3508BD517	2026-01-09	openalex.org	1	Post-Quantum Public Key Infrastructures: Hybrid Certificates, Cryptographic Combiners, and Migr...
SIG-CE5A95F7AF72	2026-01-06	openalex.org	1	PQC Planner: Post-Quantum Cryptographic Migration Planner
SIG-353070727CBB	2026-01-01	openalex.org	1	Institutional Approaches to Post-Quantum Cryptography: A Comparative Analysis of Migration Fram...
SIG-5C7568EBD975	2026-01-01	Elsevier BV	1	The Role of the Second Line of Defense in Post-Quantum Cryptography Migration and Cryptographic...
SIG-D45704585F54	2026-01-01	Elsevier BV	1	Towards a Quantum Risk Index: a risk-based framework for prioritizing Post-Quantum Cryptography...
SIG-D81DF125FECB	2026-01-01	Proceedings of the 23rd I...	1	A Capability-Driven Framework for Prioritizing Post-Quantum Cryptography Migration in Enterpris...
SIG-8BD0B85D3181	2025-12-24	openalex.org	1	Enterprise Migration to Post-Quantum Cryptography: Timeline Analysis and Strategic Frameworks
SIG-CA65D28B8AC6	2025-12-18	openalex.org	1	Harvest-Now, Decrypt-Later: A Temporal Cybersecurity Risk in the Quantum Transition
SIG-ED58CE7F1766	2025-11-24	openalex.org	1	Estimating Migration Timelines for Enterprises Transitioning to Post-Quantum Cryptography: A Re...
SIG-0C2CE2735101	2025-10-13	openalex.org	1	Post-Quantum Cryptography Migration of a Physical 5G Testbed
SIG-765D6C73A555	2026-08-13	Singtel	2	Post-quantum cryptography: what does it mean for your tech stack? - Singtel
SIG-521EDD056481	2026-06-25	R Street Institute	2	Post-Quantum Cryptography Migration in the United States: Managing Risk and Advancing Cyber Rea...
SIG-0C3636BF137F	2026-05-27	bing.com	2	The Cryptographic Migration Clock Just Got Real: A Small-Cap Just Released The Tooling Stack Fo...
SIG-47DFEDD3D7A2	2026-07-15	arxiv.org	3	Multivariate Cryptography-Based Anonymous Certificate Scheme

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When

Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:16+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:43+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.